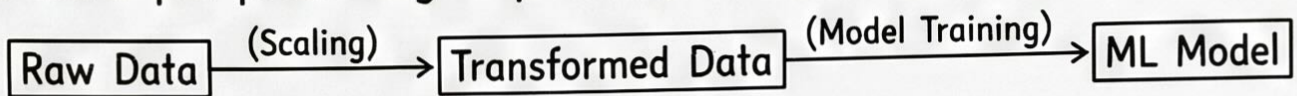


Transforming Data: Normalization & Standardization

1. What is Data Transformation?

- Process of converting raw data into a format suitable for ML models.
- Essential pre-processing step to improve model performance.



2. Normalization (Min-Max Scaling)

Purpose: Rescales features to a fixed range, typically $[0, 1]$. Helps when features have different units or scales.

Formula:
$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Range: Values are bounded between 0 and 1.

Example: Feature 'Age' (range 20-60). Let $X = 40$. $X_{\text{scaled}} = \frac{40 - 20}{60 - 20} = \frac{20}{40} = 0.5$.

3. Standardization (Z-score Scaling)

Purpose: Centers data around the mean with unit variance. Makes data distribution look Gaussian.

Properties: Mean $(\mu) \approx 0$, Standard Deviation $(\sigma) \approx 1$.

Formula:
$$X_{\text{standardized}} = \frac{X - \mu}{\sigma}$$

4. When to use Normalization vs Standardization?

• Normalization:

Use when data is NOT Gaussian or algorithm needs bounded inputs (e.g., Neural Networks, k-NN).

• Standardization:

Preferred when data follows Gaussian distribution or for algorithms assuming normality (e.g., Linear/Logistic Regression, SVMs).

5. Effect on Gradient-Based Algorithms

Effect on Gradient-Based Algorithms

- Algorithms like Gradient Descent converge faster with scaled data.
- Prevents features with larger scales from dominating the weight updates.
- Results in a more spherical error surface, allowing larger learning rates.

Feature Scales: Before vs After

